Generating HTML

With markup and DOM now firmly secured in our mind, we can turn our attention to the art of creating it from scratch. The vast majority of PHP exists to create HTML; it's what the language was created to do, and PHP is chock-full of HTML functionality. In addition to PHP's built-in tools, there is also a rich ecosystem of libraries and frameworks, as you might expect for a language that has quietly been getting on with the business of making web pages for over two decades. Of particular interest to us here are third-party tools for *templating*, which is a method of combining HTML fragments and data to assemble a whole document, and I'll be making a strong case for why you should incorporate a templating library into any decently sized project. First, let's look at the fundamentals of producing HTML from PHP.

Printing a String

The previous section had many examples of storing a full valid HTML5 document as a complete string, which we used to power `Dom\HTMLDocument` but could also just be sent directly to the client. When a web server SAPI (`cli-server`, `apache`, `fpm-fcgi`, et al.) PHP program emits data (e.g., with a `print` or `echo` statement), the emitted bytes are dispatched to the client as the response body. All you need to do is ensure that the emitted data is HTML (well-formed and valid, always!). And hey presto you have a web page. Launch a `cli-server` container to function as our web server…

```
$ docker run -it --rm \
  -p 8080:8080 \
  -v "$PWD":/usr/src/myapp \
  -w /usr/src/myapp \
  php:8.4-cli \
  php -S 0.0.0.0:8080
```

… and let's build the easiest web page you'll ever create:

```
// index.php
print <<< EOS
<!DOCTYPE html>
<html>
  <head>
```